

Semantic segmentation in remote sensing



After the survey paper [1]

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[1] M. Pastorino et al. "Segmentation sémantique en télédétection". In: *Traitement du Signal et des Images (TSI), GRETSI (2027)*. Also available as Inria research report RR-9631, <https://inria.hal.science/hal-05742224> (Sept. 2026)



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01 Introduction





From image to thematic map

- ▶ Assign to **every pixel** a label from a predefined set of semantic classes [2] [3] [4]: a problem studied for decades under the names *supervised*, *per-pixel* and *contextual* classification
- ▶ Produces a **dense map**, to be distinguished from segmentation, which partitions the image into homogeneous regions *without* attaching any meaning to them
- ▶ In remote sensing: turning satellite observations into **land-cover and land-use maps**



(a) aerial image



(b) segmentation map

Zeebruges dataset, IEEE GRSS IADF Technical Committee; imagery and ground truth: Belgian Royal Military Academy and ONERA.

[2] R. O. Duda and P. E. Hart. *Pattern Classification and Scene Analysis*. Wiley, 1973

[3] P. H. Swain and S. M. Davis. *Remote Sensing: The Quantitative Approach*. McGraw-Hill, 1978

[4] J. A. Richards. *Remote Sensing Digital Image Analysis*. Springer, 1986



What sets remote sensing apart

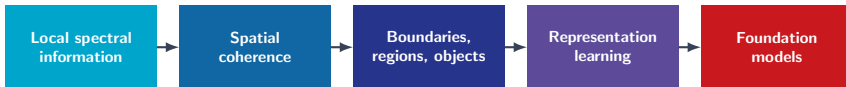
- ▶ Heterogeneous **sensors**: optical, synthetic aperture radar (SAR), LiDAR
- ▶ Widely varying **spatial resolutions**, acquisitions at **different dates**, **several modalities** combined
- ▶ **Complex geographical scenes** whose properties vary with region, season and acquisition conditions
- ▶ Consequence: the evolution of the field **is not** a mere **adaptation** of advances in **computer vision**
- ▶ It is a convergence between **statistical pattern recognition**, **signal and image processing**, **probabilistic modeling**, **mathematical morphology**, **stochastic geometry**, **object-based image analysis** and, recently, **deep learning**



Formalization and storyline

$$I : \Omega \longrightarrow \mathbb{R}^d \quad \text{then} \quad f : \Omega \longrightarrow \mathcal{C}$$

- ▶ Ω : the image domain; d : the number of variables (panchromatic band, multispectral or hyperspectral bands, SAR polarizations, LiDAR measurements, spectral indices, digital terrain models. . .)
- ▶ $\mathcal{C}$: the finite set of semantic classes
- ▶ **The storyline of this talk**: the progressive integration of increasing levels of context



02

Per-pixel statistical classification





The first approaches (1970s–1980s)

- ▶ Landsat (optical) and SeaSat (radar) create the need for automatic land-cover maps [5] [6] [7]
- ▶ Each pixel is an observation vector $\mathbf{x} = (x_1, \dots, x_d)^T$: radiometric responses, SAR intensities, indices, ancillary data [2]

$$\hat{\omega} = \arg \max_{\omega_i} P(\omega_i | \mathbf{x})$$

- ▶ Under a multivariate Gaussian assumption: the **MAP classifier**, “maximum a posteriori”, the reference method for several decades [3] [4] [8]

- [2] R. O. Duda and P. E. Hart. *Pattern Classification and Scene Analysis*. Wiley, 1973
- [3] P. H. Swain and S. M. Davis. *Remote Sensing: The Quantitative Approach*. McGraw-Hill, 1978
- [4] J. A. Richards. *Remote Sensing Digital Image Analysis*. Springer, 1986
- [5] R. Schowengerdt. *Remote Sensing: Models and Methods for Image Processing*. Academic Press, 2007
- [6] D. A. Landgrebe. *Signal Theory Methods in Multispectral Remote Sensing*. John Wiley & Sons, Ltd, 2003
- [7] F. Ulaby and D. Long. *Microwave Radar and Radiometric Remote Sensing*. 1st ed. Artech House, 2015
- [8] P. M. Mather. *Computer Processing of Remotely-Sensed Images*. Wiley, 2004



Two structural limitations

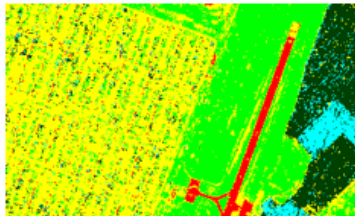
- ▶ **Curse of dimensionality** (Hughes phenomenon, described as early as 1968) [9]
 - more spectral variables $\neq$ better performance when the number of training samples remains limited
 - spectral signatures alone are not enough
- ▶ **Independence assumption on the observations**
 - each pixel is decided in isolation, without using its neighborhood
 - yet geographical objects are strongly spatially correlated: agricultural fields, forests, urban areas, road networks
- ▶ Consequence: **noisy maps**, all the more so as spatial resolution increases

[9] G. Hughes. "On the mean accuracy of statistical pattern recognizers". In: *IEEE Transactions on Information Theory* (1968)

The result: noisy maps



(a) IKONOS image, 4 m, false color



(b) per-pixel classification

IKONOS image, 4 m, 3 bands, false-color composite (near infrared, red, blue); segmentation map [13].

- The decision no longer depends on the radiometry of a single pixel alone: its **spatial environment** is needed
- The problem stops being a spectral classification and becomes one of **spatial interpretation of scenes** [10] [11] [12]

[10] S. Geman and D. Geman. "Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* (1984)

[11] J. Besag. "On the statistical analysis of dirty pictures". In: *Journal of the Royal Statistical Society: Series B* (1986)

[12] A. H. S. Solberg et al. "A Markov random field model for classification of multisource satellite imagery". In: *IEEE Transactions on Geoscience and Remote Sensing* (1996)

[13] G. Moser et al. "Land-cover mapping by Markov modeling of spatial-contextual information in very-high-resolution remote sensing images". In: *Proceedings of the IEEE* (2013)

03

Discriminative and kernel methods



Estimating the boundary rather than the distribution

- ▶ From the late 1990s: estimating **the decision boundaries directly** rather than the distribution of the observations
- ▶ A response to two difficulties: **unrealistic** Gaussian assumptions, **high dimensionality** of the data
- ▶ **Support vector machines (SVMs)**, arising from statistical learning theory [14] [15]
 - principle of **margin maximization** between classes
 - **excellent generalization** even when the number of training samples is small compared with the dimension

[14] V. N. Vapnik. *Statistical Learning Theory*. Wiley, 1998

[15] C. Cortes and V. Vapnik. "Support-vector networks". In: *Machine Learning* (1995)

Kernels: nonlinearity without losing convexity

- ▶ Kernel methods extend the framework to **nonlinear** problems while preserving a **convex formulation** [16]
- ▶ The 2000s: SVMs become one of the **most widely used** classifiers in remote sensing: multispectral, hyperspectral, multisource
- ▶ Development of **kernels tailored to geospatial observations** [17] [18]
- ▶ **But:** SVMs improve the *decision function*, not the representation: applied to observation vectors, they remain **per-pixel** classifiers
- ▶ Two research directions then evolve in parallel: *new classifiers* and *richer spatial representations*

[16] B. Schölkopf and A. J. Smola. *Learning with Kernels*. MIT Press, 2002

[17] G. Camps-Valls and L. Bruzzone. *Kernel Methods for Remote Sensing Data Analysis*. Wiley, 2009

[18] G. Mountrakis et al. "Support vector machines in remote sensing: A review". In: *ISPRS Journal of Photogrammetry and Remote Sensing* (2011)

04

Ensemble methods





Random forests

- ▶ Combining several classifiers to gain robustness and generalization: **ensemble learning**
- ▶ **Random forests**: decision trees learned on random subsets *of the observations* and *of the variables*, aggregated by majority vote [19]
- ▶ They limit overfitting while modeling nonlinear relations
- ▶ In practice: little tuning, robust to noise, handle a large number of variables, provide **variable importance** measures
- ▶ Multispectral and hyperspectral [20], then land cover, change detection, multisource data fusion [21]

[19] L. Breiman. "Random forests". In: *Machine Learning* (2001)

[20] P. O. Gislason et al. "Random forests for land cover classification". In: *Pattern Recognition Letters* (2006)

[21] M. Belgiu and L. Drăguț. "Random forest in remote sensing: A review of applications and future directions". In: *ISPRS Journal of Photogrammetry and Remote Sensing* (2016)



The pixel era: takeaways

- ▶ Discriminative and ensemble methods: clear gains **on the classifier**
- ▶ They remain **independent of the representation**, be it spectral, textural, morphological or spectral-spatial
- ▶ The fundamental limitation remains: **the spatial organization of the scene is not taken into account explicitly**
- ▶ Two complementary responses will structure what follows:
 - regularize the **labels**: contextual, Markov, variational (sections 5, 6 and 8)
 - enrich the **features**: spectral-spatial, object-based (section 7)

05

Contextual classification





Three levels of context

- ▶ As early as the 1980s and 1990s: a pixel is more likely to belong to the class of its immediate neighbors than to another one [4]
- ▶ **Local context**: interactions between neighboring pixels, small fixed-size windows [23]
- ▶ **Regional context**: texture, shape, size, homogeneity of the segmented regions; foreshadows object-based image analysis [24]
- ▶ **Global context**: overall organization of the scene, relations between objects, prior knowledge about their layout
- ▶ A structuring hierarchy that extends to contemporary deep architectures [25] [26]

[4] J. A. Richards. *Remote Sensing Digital Image Analysis*. Springer, 1986

[23] R. M. Haralick and L. G. Shapiro. "Image segmentation techniques". In: *Computer Vision, Graphics, and Image Processing* (1985)

[24] T. Blaschke. "Object based image analysis for remote sensing". In: *ISPRS Journal of Photogrammetry and Remote Sensing* (2010)

[25] R. Azencott and C. Graffigne. "Non-supervised segmentation using multi-level Markov random fields". In: *11th IAPR International Conference on Pattern Recognition, Vol. III, Conference C: Image, Speech and Signal Analysis*. 1992

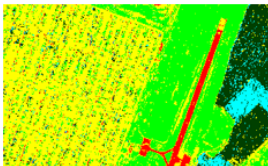
[26] G. Csurka et al. "What is a good semantic segmentation model? A survey". In: *International Journal of Computer Vision* (2023)



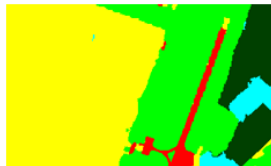
The measurable contribution of context



(a) IKONOS image



(b) per-pixel



(c) contextual

IKONOS image, 4 m, 3 bands, false-color composite (near infrared, red, blue); segmentation maps [13].

- ▶ Clear gains on multispectral, multitemporal and multisource data [12] [27] [28]
- ▶ Labels are no longer decided one by one: they must be estimated **jointly**

[12] A. H. S. Solberg et al. "A Markov random field model for classification of multisource satellite imagery". In: *IEEE Transactions on Geoscience and Remote Sensing* (1996)

[13] G. Moser et al. "Land-cover mapping by Markov modeling of spatial-contextual information in very-high-resolution remote sensing images". In: *Proceedings of the IEEE* (2013)

[27] L. Bruzzone and S. B. Serpico. "An iterative technique for the detection of land-cover transitions in multitemporal remote-sensing images". In: *IEEE Transactions on Geoscience and Remote Sensing* (1997)

[28] L. Bruzzone and S. B. Serpico. "A neural-statistical approach to multitemporal and multisource remote-sensing image classification". In: *IEEE Transactions on Geoscience and Remote Sensing* (1999)

06

Markov and Bayesian models





Markov random fields: the first formalization

- ▶ Labels are no longer independent: they are random variables defined on a **neighborhood graph** [10] [11] [29]
- ▶ The Markov assumptions *and* the equivalence between Markov random fields and Gibbs distributions

$$\hat{Y} = \arg \max_Y P(Y | X) \quad \Longleftrightarrow \quad \hat{Y} = \arg \min_Y U(Y | X)$$

- ▶ X : the observations; Y : the labels to be estimated
- ▶ Bayes' theorem turns MAP estimation into an **energy minimization**

[10] S. Geman and D. Geman. "Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* (1984)

[11] J. Besag. "On the statistical analysis of dirty pictures". In: *Journal of the Royal Statistical Society: Series B* (1986)

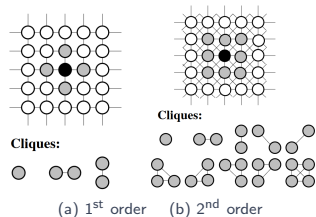
[29] Z. Kato and J. Zerubia. "Markov random fields in image segmentation". In: *Foundations and Trends in Signal Processing* (2012)



Data term + regularization

$$U(Y | X) = U_d(Y | X) + \beta U_r(Y | X)$$

- ▶ U_d : the **data term**, derived from the likelihood
- ▶ U_r : **spatial regularization**, promoting consistency between neighboring pixels
- ▶ β : the trade-off between fidelity and regularity
- ▶ 1st- or 2nd-order neighborhoods; clique potentials **penalize class changes** between **adjacent pixels** [29]



Pixel neighborhoods on a grid and the associated cliques [29].

[29] Z. Kato and J. Zerubia. "Markov random fields in image segmentation". In: *Foundations and Trends in Signal Processing* (2012)



Optimizing: a hard problem

- ▶ Finding the optimal configuration is a hard combinatorial optimization problem $\Rightarrow$ **suboptimal** strategies
- ▶ **Iterated conditional modes (ICM)**, **simulated annealing** [30], the **Gibbs sampler** [10]
- ▶ **Graph cuts** [31], **loopy belief propagation** [32]
- ▶ Application areas: spatial consistency of classifications, multisource segmentation, restoration, change detection, optical / radar / LiDAR fusion [12] [33] [34]
- ▶ Beyond these applications: a **general probabilistic formulation of spatial regularization**

[10] S. Geman and D. Geman. "Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* (1984)

[12] A. H. S. Solberg et al. "A Markov random field model for classification of multisource satellite imagery". In: *IEEE Transactions on Geoscience and Remote Sensing* (1996)

[30] S. Kirkpatrick. "Optimization by simulated annealing: Quantitative studies". In: *Journal of Statistical Physics* (1984)

[31] Y. Boykov et al. "Fast approximate energy minimization via graph cuts". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2001)

[32] K. Tanaka et al. "Loopy belief propagation and probabilistic image processing". In: *2003 IEEE XIII Workshop on Neural Networks for Signal Processing*. 2003

[33] H. Derin and H. Elliott. "Modeling and segmentation of noisy and textured images using Gibbs random fields". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* (1987)

[34] G. Moser and S. B. Serpico. "Combining support vector machines and Markov random fields in an integrated framework for contextual image classification". In: *IEEE Transactions on Geoscience and Remote Sensing* (2013)



From MRFs to CRFs

- ▶ Conditional random fields model $P(Y | X)$ directly rather than the joint distribution [35]: this makes **complex discriminative features** easier to use
- ▶ **Fully connected CRFs**: better preserved object boundaries [36]
- ▶ Integration into deep networks: refinement modules, **differentiable layers** [37]; hierarchical Markov frameworks and learned CRFs [38] [39] [40] [41]
- ▶ **The idea that runs through the field**: a map must be consistent both with the local observations *and* with the spatial organization of the scene

- [35] J. Lafferty et al. "Conditional random fields: Probabilistic models for segmenting and labeling sequence data". In: *International Conference on Machine Learning (ICML)*. 2001
- [36] P. Krähenbühl and V. Koltun. "Efficient inference in fully connected CRFs with Gaussian edge potentials". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2011
- [37] S. Zheng et al. "Conditional random fields as recurrent neural networks". In: *IEEE International Conference on Computer Vision (ICCV)*. 2015
- [38] M. Pastorino et al. "Multisensor and multiresolution remote sensing image classification through a causal hierarchical Markov framework and decision tree ensembles". In: *Remote Sensing* (2021)
- [39] M. Pastorino et al. "Semantic segmentation of remote-sensing images through fully convolutional neural networks and hierarchical probabilistic graphical models". In: *IEEE Transactions on Geoscience and Remote Sensing* (2022)
- [40] M. Pastorino et al. "CRFNet: A deep convolutional network to learn the potentials of a CRF for the semantic segmentation of remote sensing images". In: *IEEE Transactions on Geoscience and Remote Sensing* (2024)
- [41] A. Voisin et al. "Classification of very high resolution SAR images of urban areas using copulas and texture in a hierarchical Markov random field model". In: *IEEE Geoscience and Remote Sensing Letters* (2013)

07

Spectral-spatial methods





Enriching the representation, not the labels

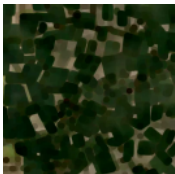
- ▶ Contextual models regularize the **labels**; here the **features** are enriched *before* classification [6]
- ▶ Ideas: the **increasing resolution** of sensors, both spatial and spectral
 - a single class exhibits strong spectral variability
 - different objects sometimes have very similar signatures
- ▶ Texture, shape, size, spatial organization and neighborhood relations become **as discriminative as the spectral values themselves** [42]

[6] D. A. Landgrebe. *Signal Theory Methods in Multispectral Remote Sensing*. John Wiley & Sons, Ltd, 2003

[42] S. Dellepiane et al. "SAR image recognition by integration of intensity and textural information". In: *International Journal of Remote Sensing* (1991)



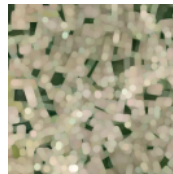
Mathematical morphology



erosion



original RGB image



dilation

Erosion (left) and dilation (right) of the original RGB satellite image (center).

- ▶ Local statistics and textural features [22]
- ▶ **Mathematical morphology** operators: erosion, dilation and their compositions [43] [44]

[22] R. M. Haralick et al. "Textural features for image classification". In: *IEEE Transactions on Systems, Man, and Cybernetics* (1973)

[43] J. Serra. *Image Analysis and Mathematical Morphology*. Academic Press, 1982

[44] P. Soille. *Morphological Image Analysis*. Springer, 2003



Morphological profiles and attribute profiles

- ▶ **Extended morphological profiles (EMPs)** [45] then **attribute profiles** [46]: describing spectral properties and spatial structure simultaneously **at several scales**
- ▶ Reference methods for hyperspectral data and very high resolution
- ▶ Combined with SVMs: remarkable performance [17] [47], but the principle is general (random forests, neural networks. . .)
- ▶ Dimensionality reduction, subspace-based methods, multisource data fusion [48] [49]
- ▶ Using **segmented regions** rather than pixels as analysis units: greater spatial consistency, less noise [50]

[17] G. Camps-Valls and L. Bruzzone. *Kernel Methods for Remote Sensing Data Analysis*. Wiley, 2009

[45] J. A. Benediktsson et al. "Classification of hyperspectral data from urban areas based on extended morphological profiles". In: *IEEE Transactions on Geoscience and Remote Sensing* (2005)

[46] M. Dalla Mura et al. "Morphological attribute profiles for the analysis of very high resolution images". In: *IEEE Transactions on Geoscience and Remote Sensing* (2010)

[47] M. Fauvel et al. "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles". In: *IEEE Transactions on Geoscience and Remote Sensing* (2008)

[48] A. Plaza et al. "Recent advances in techniques for hyperspectral image processing". In: *Remote Sensing of Environment* (2009)

[49] G. Camps-Valls et al. "Advances in hyperspectral image classification". In: *IEEE Signal Processing Magazine* (2014)

[50] Y. Tarabalka et al. "Spectral-spatial classification of hyperspectral imagery based on partitional clustering techniques". In: *IEEE Transactions on Geoscience and Remote Sensing* (2009)

08

Variational methods





Segmenting by minimizing an energy

- ▶ Late 1980s, in parallel with Markov random fields: **no explicit probability distribution on the labels**, but the **search for the partition** achieving the best trade-off between **fidelity to the observations** and **regularity** [29] [51] [52]
- ▶ **Active contours**: boundaries are deformable curves evolving under the action [52] [53]
 - of *internal forces* (contour regularity)
 - of *external forces* (derived from the image information)
- ▶ An **explicit** representation of contours, and the starting point of a large family of energy-minimization methods

[29] Z. Kato and J. Zerubia. "Markov random fields in image segmentation". In: *Foundations and Trends in Signal Processing* (2012)

[51] D. Mumford and J. Shah. "Optimal approximations by piecewise smooth functions and associated variational problems". In: *Communications on Pure and Applied Mathematics* (1989)

[52] C. Samson et al. "A variational model for image classification and restoration". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2000)

[53] M. Kass et al. "Snakes: Active contour models". In: *International Journal of Computer Vision* (1988)



The Mumford–Shah functional

$$E(u, K) = \underbrace{\int_{\Omega} (u - I)^2 dx}_{\text{data fidelity}} + \mu \underbrace{\int_{\Omega \setminus K} |\nabla u|^2 dx}_{\text{within-region smoothness}} + \nu \underbrace{|K|}_{\text{boundary complexity}}$$

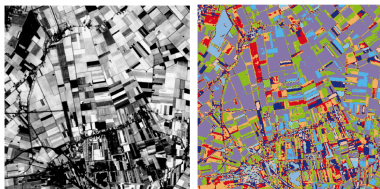
- ▶ u : the reconstructed image, I : the observed image, K : the set of discontinuities (the boundaries) [51]
- ▶ $\mu > 0$: smoothness of the solution inside the regions; $\nu > 0$: weight of the boundary measure
- ▶ An explicit trade-off between **data fit** and **regularity of the solution**, a principle shared by many segmentation methods
- ▶ One of the most influential variational models in image processing

[51] D. Mumford and J. Shah. "Optimal approximations by piecewise smooth functions and associated variational problems". In: *Communications on Pure and Applied Mathematics* (1989)



Numerical robustness and applications

- ▶ **Level sets**: contours represented implicitly, topology changes handled automatically [54]
- ▶ **Region-based models**: region statistics rather than gradients alone, robust to noise and low contrast [55]
- ▶ In remote sensing: **high-resolution optical segmentation, restoration, fusion, object extraction; classification and restoration** unified in a single functional [52]
- ▶ **Convergence**: Bayesian, Markov random field, variational and graph-based formulations are different expressions of the same principle: **data term + regularization**



(a) SPOT image (b) segmentation
(a) SPOT satellite image © CNES; (b) segmentation by the variational method of [52].

[52] C. Samson et al. "A variational model for image classification and restoration". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2000)

[54] S. Osher and J. A. Sethian. "Fronts propagating with curvature-dependent speed: Algorithms based on Hamilton–Jacobi formulations". In: *Journal of Computational Physics* (1988)

[55] T. F. Chan and L. A. Vese. "Active contours without edges". In: *IEEE Transactions on Image Processing* (2001)

09

Deep learning





2012, the turning point

- ▶ **AlexNet** demonstrates a dramatic improvement in natural-image classification [56]
- ▶ **Learning hierarchical representations** directly from the data, without hand-crafted design of spectral, textural or geometric features
- ▶ Performance now depends on the ability of the network to learn **jointly** a relevant representation *and* the decision function
- ▶ Rapid adoption in remote sensing: more discriminative representations, **transfer** of models pretrained on large datasets of natural images [57] [58]
- ▶ **Spectral-spatial approaches** shift from *hand-crafted* to *learned* representations

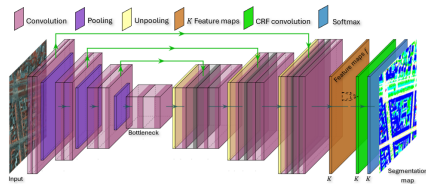
[56] A. Krizhevsky et al. "ImageNet classification with deep convolutional neural networks". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2012

[57] Y. Chen et al. "Deep learning-based classification of hyperspectral data". In: *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (2014)

[58] M. Castelluccio et al. "Land use classification in remote sensing images by convolutional neural networks". In: *arXiv preprint arXiv:1508.00092* (2015)

Dense prediction: FCNs and encoder–decoder

- **FCNs**: fully connected layers replaced by convolutional operations $\Rightarrow$ dense, pixel-level prediction, learned end-to-end [59]
- **U-Net**: an encoder–decoder whose *skip connections* restore the localization lost in downsampling, hence better boundaries and smaller objects [61]
- **Applications**: land cover, buildings, roads, hydrographic networks [62] [63] [64]



FCN architecture

Applied to the aerial images of the ISPRS 2D Semantic Labeling Challenge dataset (Potsdam) [60].

- [59] J. Long et al. "Fully convolutional networks for semantic segmentation". In: *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2015
- [60] M. Pastorino et al. "Probabilistic graphical models meet deep learning for semantic segmentation: Mathematical connections and recent developments". In: *IEEE Signal Processing Magazine* (2026)
- [61] O. Ronneberger et al. "U-Net: Convolutional networks for biomedical image segmentation". In: *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2015
- [62] N. Audebert et al. "Beyond RGB: Very high resolution urban remote sensing with multimodal deep networks". In: *ISPRS Journal of Photogrammetry and Remote Sensing* (2018)
- [63] Z. Zhang et al. "Road extraction by deep residual U-Net". In: *IEEE Geoscience and Remote Sensing Letters* (2018)
- [64] F. Isikdogan et al. "Surface water mapping by deep learning". In: *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (2017)



Context, scales, modalities

- ▶ **Residual networks**: learning deeper models while limiting the vanishing gradient [65]
- ▶ **DeepLab**: dilated convolutions and multiscale aggregation (Atrous Spatial Pyramid Pooling, ASPP) to enlarge the receptive field *without* degrading the resolution of the predictions [66] [67]
- ▶ **Feature pyramids**: objects of widely varying sizes [68], decisive in remote sensing where a single scene contains vehicles, buildings, fields and forest stands
- ▶ **Multimodality**: learning shared representations from optical, SAR and LiDAR data, digital terrain models and ancillary geospatial data [62] [69]

- [62] N. Audebert et al. "Beyond RGB: Very high resolution urban remote sensing with multimodal deep networks". In: *ISPRS Journal of Photogrammetry and Remote Sensing* (2018)
- [65] K. He et al. "Deep residual learning for image recognition". In: *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016
- [66] L.-C. Chen et al. "Rethinking atrous convolution for semantic image segmentation". In: *arXiv preprint arXiv:1706.05587* (2017)
- [67] L.-C. Chen et al. "Encoder-decoder with atrous separable convolution for semantic image segmentation". In: *European Conference on Computer Vision (ECCV)*. 2018
- [68] T.-Y. Lin et al. "Feature pyramid networks for object detection". In: *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2017
- [69] L. Ma et al. "Deep learning in remote sensing applications: A meta-analysis and review". In: *ISPRS Journal of Photogrammetry and Remote Sensing* (2019)



What deep learning does not eliminate

- ▶ Spatial context, multiscale representation, data fusion, object hierarchy: **reformulated** within an end-to-end learning framework, not abandoned
- ▶ Persistent limitations of convolutional networks:
 - pixel-level annotations are **costly**, particularly in remote sensing
 - limited **generalization** across sensors, geographical regions and acquisition modalities
 - difficulty in modeling **very long-range** spatial dependencies
- ▶ These limitations pave the way for **attention mechanisms**, and then for **geospatial foundation models**

10

Transformers and foundation models





Attention in segmentation

- ▶ Developed for language processing, transformers directly model the relations between **distant elements** of a sequence [70]; their adaptation to vision opens a new phase [71]
- ▶ **Swin Transformer** [72] and **SegFormer** [73]: combining a *hierarchical* representation of the image with a *global* modeling of the context
- ▶ Particularly relevant in remote sensing: identifying a building, a road or a field does not rely on local appearance alone, but on its **organization within the scene**

[70] A. Vaswani et al. "Attention is all you need". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2017

[71] A. Dosovitskiy et al. "An image is worth 16x16 words: Transformers for image recognition at scale". In: *International Conference on Learning Representations (ICLR)*. 2021

[72] Z. Liu et al. "Swin transformer: Hierarchical vision transformer using shifted windows". In: *IEEE International Conference on Computer Vision (ICCV)*. 2021

[73] E. Xie et al. "SegFormer: Simple and efficient design for semantic segmentation with transformers". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2021



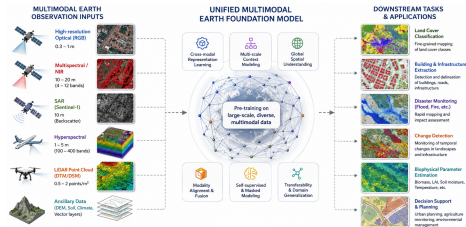
Learning without annotations

- ▶ **Self-supervised learning** becomes a major research direction
- ▶ **Masked autoencoders** show that rich visual representations can be learned from large amounts of **unlabeled** data [74]
- ▶ A direct response to an **imbalance** specific to the field:
 - satellite archives are **abundant**
 - pixel-level annotations remain **scarce and costly**

[74] K. He et al. "Masked autoencoders are scalable vision learners". In: *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022

Geospatial foundation models

- **Pretrained on very large volumes of data**, then adapted: classification, segmentation, object detection and change detection [75] [76] [77] [78] [79]
- The goal: **generic representations**, **transferable** across sensors, geographical regions and applications
- The **cost** of pretraining (compute, energy) $\Rightarrow$ environmental footprint and accessibility
- Not a break: the same questions (heterogeneity, context, generalization) are now **learned** rather than **hand-crafted**



general principle of a geospatial foundation model

- [75] C. Brown et al. "AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data". In: *arXiv preprint arXiv:2507.22291* (2025)
- [76] Y. Cong et al. "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2022
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11

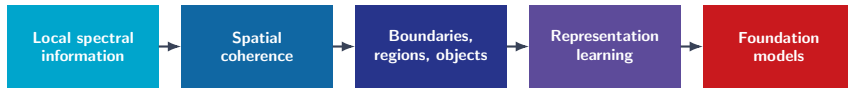
Conclusion





A nonlinear history

- ▶ Not a simple trajectory from statistical classifiers to deep networks, but the **convergence of several families of methods**: statistical classification, kernel methods, ensemble methods, contextual models, Markov random fields, variational methods, stochastic geometry, deep learning, foundation models
- ▶ One unifying reading: the **progressive integration of increasing levels of context**



- ▶ One recurring principle: **data term + regularization**, from Markov random fields to the loss functions of deep networks

Outlook

- ▶ Geographic generalization
- ▶ Lack of annotations
- ▶ Optical, radar and LiDAR fusion
- ▶ Accounting for the temporal dimension
- ▶ Model interpretability
- ▶ Operational robustness
- ▶ Semantic segmentation in remote sensing remains an **open field**, at the intersection of signal and image processing, probabilistic modeling, geometry, computer vision and artificial intelligence, in a geospatial setting

Thank you for your attention.



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Figure credits

- ▶ Aerial image and segmentation map: *Zeebruges* dataset, IEEE GRSS Image Analysis and Data Fusion (IADF) Technical Committee; imagery and ground-truth data provided by the Belgian Royal Military Academy and ONERA (the French Aerospace Lab).
- ▶ IKONOS satellite image, 4 m, 3 bands, false-color composite (near infrared, red, blue); per-pixel and contextual segmentation maps [13].
- ▶ First- and second-order neighborhoods on a grid, and the associated cliques [29].
- ▶ Erosion and dilation of an RGB satellite image.
- ▶ SPOT satellite image © CNES; segmentation map obtained with the variational method of [52].
- ▶ Architecture of an FCN applied to the aerial images of the ISPRS *2D Semantic Labeling Challenge* dataset (Potsdam) [60].
- ▶ General principle of a geospatial foundation model.

[13] G. Moser et al. "Land-cover mapping by Markov modeling of spatial-contextual information in very-high-resolution remote sensing images". In: *Proceedings of the IEEE* (2013)

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